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# Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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## Abstract

Generative image models trained on large-scale image-text corpora now synthesize highly realistic images, yet they continue to omit, suppress, or distort concepts that should plausibly be represented in their training data, including human hands and objects appearing in groups of four. Because these failures have primarily been documented through anecdotes or narrowly specified probes, the field still lacks a systematic account of whether such errors are isolated prompt-level curiosities or symptoms of broader distributional mismatch. We address this gap by formalizing and measuring *conceptual blindspots*: concepts present in natural data that are systematically absent, suppressed, or misrepresented in generated images. Our method uses sparse autoencoders (SAEs) to recover interpretable concept embeddings, enabling direct quantitative comparison of concept prevalence between real and synthesized images. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—and use this representation to analyze conceptual disparities at both distribution and datapoint levels. Applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the framework identifies suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the individual-datapoint level, the same representation isolates memorization artifacts, including cases in which models reproduce highly specific visual templates seen during training. Together, these results establish a theoretically grounded and empirically scalable framework for evaluating the conceptual fidelity of generative image models relative to an underlying data-generating process.

## 1 Introduction

Generative image models trained on large-scale datasets now provide sufficient visual competence to support applications across computer vision [1-9] and adjacent domains [9-15]. This progress nevertheless coexists with a persistent, empirically documented limitation: qualitative [16-18] and quantitative studies [19-20] show that models can fail on concepts that appear simple from a human perspective, including human hands [21-24], objects appearing in groups of four [25], and negations or object relations [20, 26]. When prompted with such concepts, models often synthesize adjacent visual structure rather than the intended concept itself—for example, a hand-like object with six fingers. We refer to these failures as “conceptual blindspots”<sup>1</sup>: concepts expected to be represented in the training distribution but absent, suppressed, or distorted in model outputs. The central question is therefore no longer whether individual prompts can expose surprising errors, but whether those errors instantiate a measurable phenomenon in which generative models under-produce or over-produce broad sets of concepts relative to the natural data distribution.

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<sup>1</sup>We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.



Figure 1:  $\delta(k)$  quantifies the prevalence of concept  $c_k$  in generated images relative to its frequency in natural data. Concepts with  $\delta(k) < 0.1$  are classified as **suppressed conceptual blindspots**, whereas concepts with  $\delta(k) > 0.9$  are classified as **exaggerated conceptual blindspots**. Images from four popular generative image models illustrate both blindspot regimes and comparatively aligned concepts.

Existing evaluation protocols do not directly answer this concept-level question for generative image models [28, 29]. Widely used metrics such as FID [30] summarize image realism or distributional proximity at a coarse scale, and can therefore obscure failures concentrated in particular concepts. Related measures based on CLIPScore, generation diversity, anomaly detection, or distribution coverage [31–33] provide useful aggregate evidence, but they do not operate at the granularity of fine-grained visual features or concepts [34, 35]. Conversely, qualitative evaluations such as participant studies [5, 17, 36–38] and open-ended model exploration [39] can expose concrete failures, but they do not provide a scalable, repeatable, and model-comparable procedure for discovering conceptual blindspots.

**This work.** We develop an automated, unsupervised methodology for identifying concepts whose prevalence differs between a natural data-generating process and a trained generative image model. The paper makes three connected contributions that link the formal object of study, the technical estimator, and the empirical evidence:

- **Formalizing Conceptual Blindspots in Generative Image Models.** We define conceptual blindspots as concept-level discrepancies between natural and generated images, yielding a quantitative framework for measuring suppression and exaggeration relative to the data-generating process (Section 3). This formulation moves evaluation beyond anecdotal failures and hand-selected probes toward a principled distributional comparison.
- **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse Autoencoders.** We instantiate the formalization with sparse autoencoders (SAEs), which provide a shared, interpretable concept basis for comparing natural and synthesized images (Section 4). We train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts, the largest such RA-SAE to date.
- **Exploratory Tool and Analysis.** We provide an exploratory web tool for distribution- and datapoint-level analysis of blindspots across models (Sections 5.1–5.4). Applying the method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5), we identify suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), as illustrated in Fig. 1.

The model, code, and web tool are available at a software repository.

## 2 Related Work

### 2.1 Explainability in Vision

Early explainable-AI work in computer vision emphasized attribution methods that localize influential input regions [40, 42]. Although useful for saliency analysis, these methods provide limited semantic access to learned representations and can yield unreliable explanations [43–45]. Concept-based interpretability [46] supplies a more appropriate level of abstraction by identifying semantically meaningful directions in neural networks, thereby exposing not only where a model attends but also which concepts and visual structures mediate its behavior [47–49].

Recent analyses show that prominent concept-based methods—ACE [44], CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—can be formulated as variants of a dictionary-learning problem with



different constraints [52]. Within this family, SAEs provide a scalable mechanism for extracting sparse, interpretable concept activations. Although standard SAEs have known limitations, including overly specific features [53], compositionality challenges [54], and limited intervention effects [55], improved variants such as archetypal SAE (RA-SAE) [56] are designed to yield more stable and data-faithful dictionaries.

Other interpretability tools, including prompt-based probing [57], attention or activation visualizations [58, 59], and dataset-level statistics [32, 33], provide complementary but partial evidence. They generally operate on complete images, prompts, or aggregate coverage measures rather than on fine-grained features and concepts [34, 35]. They also often require subjective interpretation and do not separate distinct failure modes [60]. These limitations motivate an explicitly concept-level diagnostic for generative image models [28], one that preserves semantic granularity while scaling across many concepts and model families.

## 2.2 Generative Image Models

Diffusion-based methods now occupy a central position in generative vision modeling across images [4, 61–64], video [65–72], and 3D generation [7, 73–75]. In image generation, this trajectory begins with denoising diffusion probabilistic models (DDPMs) [76] and extends through non-Markovian denoising diffusion implicit models (DDIMs) [77].

The Stable Diffusion (SD) [9] family made diffusion models broadly accessible to research and open-source communities. The original SD model was followed by SD 2 [78], SD 3 [79], and SD XL [80], together with extensions that condition generation on style [81, 82], pose [83], identity [84, 85], and alternative input modalities such as image-to-image generation. Additional latent diffusion systems, including Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89], broaden the architectural landscape in which concept-level comparisons are scientifically necessary.

## 2.3 Datasets for Generative Image Models

The performance of contemporary generative vision models depends substantially on computational scale and large internet-derived datasets [90–92]. LAION-5B [93], collected from Common Crawl [94], has been especially influential for open-source text-to-image systems, including SD and its derivatives; it contains more than 5B image-caption pairs, of which 2.3B are in English. Other widely used resources include COYO-700M [95] and Conceptual Captions [96], with 700M and 3M image-caption pairs, respectively.

The adoption of LAION-5B also intensified concerns about bias and unsafe content [97–101]. Despite filtering, harmful material remained [97–99], including NSFW images [100] and hundreds of CSAM instances [101], prompting temporary removal from official channels. LAION-5B further contains low-quality images [102] and internet-style captions, such as product descriptions, that can differ from user prompts supplied to trained models [103]. These dataset properties make concept-level comparison between natural and generated images especially important: aggregate quality metrics can miss systematic distortions inherited from, or amplified relative to, the training distribution.

## 2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning constructs sparse representations in which each sample is reconstructed from a small number of dictionary atoms [104, 106]. Building on compressed sensing theory [107, 108], the area has progressed from vector quantization [109] to methods such as Non-negative Matrix Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. More recent developments include online learning [114], structured sparsity [115], theoretical recovery guarantees [116, 117], and increasingly direct connections to deep representation learning [118, 119].

Sparse coding has also been used to study the emergence of high-level concepts inside diffusion models [120]. Before diffusion models, concept-grounded interpretability for generative systems relied on concept-bottleneck models, which require human intervention during training [121], or post-hoc detectors that retrofit concept supervision [122]. Because both strategies depend on human-defined concepts, they can miss systematic concept-level trends that are not specified in advance.

Aggregate metrics, including precision, recall, density, and coverage [35, 123], as well as latent density scores that estimate sample quality from the model’s latent space [124], are useful for



measuring global generative behavior. They are less suited to identifying which specific concepts are under- or over-represented. This gap motivates a scalable, unsupervised framework that quantifies concept-level failure modes in generative image models [125] and links those measurements to interpretable visual structure.

### 3 Formalizing Conceptual Blindspots in Generative Models

We formalize *conceptual blindspots* as systematic discrepancies between the concepts expressed in natural images and those expressed in model-generated images. The resulting framework assigns quantitative meaning to concepts that a generative model suppresses or exaggerates relative to the data distribution. The construction is summarized in Fig. 2. Although the definitions rely on standard assumptions from representation learning and identifiability [126–130], our empirical analysis shows that the induced measurements expose meaningful model behavior even when these assumptions hold only approximately.

**Definition 1 (Data-Generating Process).** Let  $\mathcal{C} \subset \mathbb{R}^K$  denote a latent space with a Boltzmann prior  $p(c) = \exp(-E(c))Z^{-1}$ , where  $E(\cdot)$  denotes an energy function that linearly decomposes over individual latents and  $Z$  is the corresponding partition function, i.e.,  $E(c) = \sum_k E(c_k)$  and hence  $p(c) = \prod_k p_k(c_k)$ . A data-generating process (DGP) is an invertible function  $G : \mathcal{C} \rightarrow \mathcal{X}$  that maps the latents  $c \in \mathcal{C}$  to observations  $x \in \mathcal{X}$ , i.e.,  $x = G(c)$ .

For notational simplicity, we use  $p(\cdot)$  to denote both the latent density  $p(c)$  and its push-forward to image space  $p(x)$ , where  $x = G(c)$ . The invertibility of  $G$  justifies this convention because it induces a valid distribution over  $\mathcal{X}$  via the change-of-variables formula. Accordingly, the coordinates of the latent space define the **Concepts** that organize the image distribution  $P_{\mathcal{X}}$  over the observation space  $\mathcal{X}$ . Such latents may encode attributes including color, shape, size, and location [131][132]. We further allow the data-generating process to associate each image with a text description  $t \in \mathcal{T}$ , while leaving the captioning mechanism implicit. A text-conditioned **Generative image model**  $g_{\theta}$  with parameters  $\theta$  is then trained on image–text pairs to map a noise signal  $\eta \sim \mathcal{N}(0, I)$  and a scene description  $t$  to an image  $x$  depicting that description.

To identify blindspots in  $g_{\theta}$ , we compare the probability mass assigned to each concept by the data-generating process with the corresponding probability mass induced by the model. Starting from an evaluation set of natural image–text pairs  $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$ , we construct a synthetic set  $\mathcal{D}'_{\mathcal{X}}$  by generating images from the same text descriptions. Concept occurrence is estimated with an **Energy model**  $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$  that maps each image to a  $d$ -dimensional vector whose  $k^{\text{th}}$  coordinate represents the energy of concept  $k$ , i.e.,  $\xi_k(x) = E(c_k)$ . The invertibility assumption on the data-generating process makes these estimates well defined. The probability associated with an image  $x$  can then be expressed as  $p_k(x) \propto \exp(-\sum_k \xi_k(x))$  (where  $Z_{\xi}$  is the partition function), yielding the population-level estimate  $p_k(\mathcal{D}_{\mathcal{X}}) = \prod_{x \in \mathcal{D}_{\mathcal{X}}} p_k(x)$ . Using this construction and the sigmoid map  $\sigma(\cdot)$ , we define the average energy difference for concept  $k$  between  $\mathcal{D}_{\mathcal{X}}$  and  $\mathcal{D}'_{\mathcal{X}}$ .

**Definition 2 (Energy Difference).** Let  $x \in \mathcal{D}_{\mathcal{X}}$  denote a real image sampled from the data-generating process  $G$ , and let  $x' \in \mathcal{D}'_{\mathcal{X}}$  be a synthetic image generated by the model  $g_{\theta}$ . Let  $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$  denote the energy assigned to the  $k^{\text{th}}$  concept by the energy model  $\xi$ . We define the

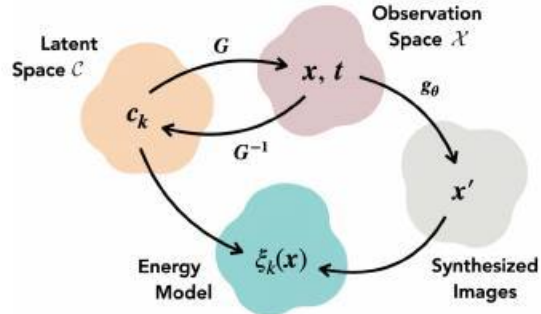


Figure 2: **Intuitive mapping of our framework.** Latent concepts  $c_k \in \mathcal{C}$  are mapped to observations  $(x, t)$  through the (invertible) DGP.  $S_{\theta}$  generates images  $x'$  from  $t$ . The energy model  $\xi_k(x)$  extracts concept representations from both  $x$  and  $x'$ , enabling concept-distribution comparisons that identify suppressed and exaggerated blindspots.



178 energy difference for concept  $k$  as:

$$\begin{aligned}\delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_{\mathcal{X}})}{p_k(\mathcal{D}_{\mathcal{X}}) + p_k(\mathcal{D}'_{\mathcal{X}})},\end{aligned}\tag{1}$$

179 where the expectations are taken over  $\mathcal{D}'_{\mathcal{X}}$  and  $\mathcal{D}_{\mathcal{X}}$ , respectively, and  $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$   
180 denotes the unnormalized conceptual probability mass of dataset  $\mathcal{D}$  under concept  $k$ .

181 Thus, the energy difference for concept  $k$  expresses the relative probability of observing that concept  
182 in a target set of observations (here,  $\mathcal{D}'_{\mathcal{X}}$ ) compared with a baseline set (here,  $\mathcal{D}_{\mathcal{X}}$ ). This statistic  
183 directly induces the operational definition of conceptual blindspots.

184 **Definition 3 (Suppressed / Exaggerated Conceptual Blindspots).** Given a generative image model  
185  $g_\theta$ , we say, compared to the data-generating process  $G$ ,  $c_k$  is a suppressed conceptual blindspot in  
186 the model if  $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$  and exaggerated if  $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$ .

187 A conceptual blindspot is therefore a concept whose likelihood in generated images deviates markedly  
188 from its prevalence under the data-generating process, either through suppression or exaggeration.  
189 Suppressed concepts have disproportionately low activation (e.g.,  $\delta(k) < \lambda_{\min}$ ), whereas exaggerated  
190 concepts are overrepresented (e.g.,  $\delta(k) > \lambda_{\max}$ ). Throughout the empirical analysis, we use  
191  $\lambda_{\min} = 0.1$  and  $\lambda_{\max} = 0.9$  to isolate these two high-confidence regimes.

192 This definition is related to “mode collapse” in prior work (e.g., see Bau et al. [39]), but it shifts the  
193 unit of analysis. Mode collapse concerns the exaggerated or suppressed probability of generating  
194 entire images; conceptual blindspots instead concern altered probabilities of specific concepts within  
195 images. For instance, if a model fails to generate objects with a white background, the missing  
196 background attribute is a suppressed conceptual blindspot rather than a missing image mode.

## 197 4 Method: Operationalizing the Definition of Conceptual Blindspots

198 We now instantiate the formal definition as a concrete pipeline for detecting conceptual blindspots in  
199 a generative model  $g_\theta$ . Sec. 3 requires two empirical objects: (i) generated images sampled from  $g_\theta$   
200 under captions that support direct comparison with the natural data-generating process, and (ii) an  
201 energy model that maps both natural and generated images into a common concept basis. We use  
202  $\|\cdot\|_F$  for the Frobenius norm and  $\|\cdot\|_0$  for the number of nonzero entries (the  $\ell_0$  pseudo-norm).  
203 For a vector or matrix  $\mathbf{X}$ ,  $\mathbf{X} \geq 0$  denotes element-wise non-negativity. For  $n > 0$ ,  $[n] := \{1, \dots, n\}$ ,  
204 and the  $i$ -th row of a matrix  $\mathbf{A}$  is denoted by  $\mathbf{A}_i$ .

205 **From Prompts to Latent Representations.** To identify conceptual blindspots in  $g_\theta$ , we construct  
206 prompt-matched comparisons between natural image-caption pairs  $(\mathbf{x}, t)$  sampled from the data-  
207 generating process  $G$  and synthetic counterparts generated by  $g_\theta$  from the same text descriptions.

208 Given a caption  $t$ , we synthesize the paired image  $\mathbf{x}'$  with a pretrained text-to-image generator  
209  $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$  implemented as a denoising diffusion probabilistic model (DDPM) [78] [86] [87].  
210 Sampling proceeds in latent space through the reverse trajectory  $(\gamma_t)_{t=0}^T$ :

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

211 where  $\alpha_t \in (0, 1)$  and  $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$  follow the standard cosine noise schedule. The final latent  
212  $\gamma_0$  is decoded by a pretrained VAE to obtain  $\mathbf{x}' = \text{VAE}(\gamma_0)$ . In the analyses below,  $g_\theta$  is treated as  
213 a black-box prompt-to-image map  $t \mapsto \mathbf{x}'$ , so the blindspot measurements depend only on paired  
214 inputs and outputs rather than on model-internal access.

215 **Defining the Energy Model.** Prior work shows that self-supervised learning can recover structure  
216 related to the data-generating process and identify the underlying energy function up to linear  
217 transformations [126] [128] [133] [134]. We therefore use DINOv2 [135], a state-of-the-art self-  
218 supervised vision model, as the feature extractor for the energy model. Because the number of  
219 concepts underlying the DGP may exceed the dimensionality of the model feature space [51] [136],  
220 we train sparse autoencoders (SAEs) on DINOv2 features to recover subspaces associated with these  
221 concepts [50] [56] [137] [139]. The operational assumption is that if DINOv2 represents concepts



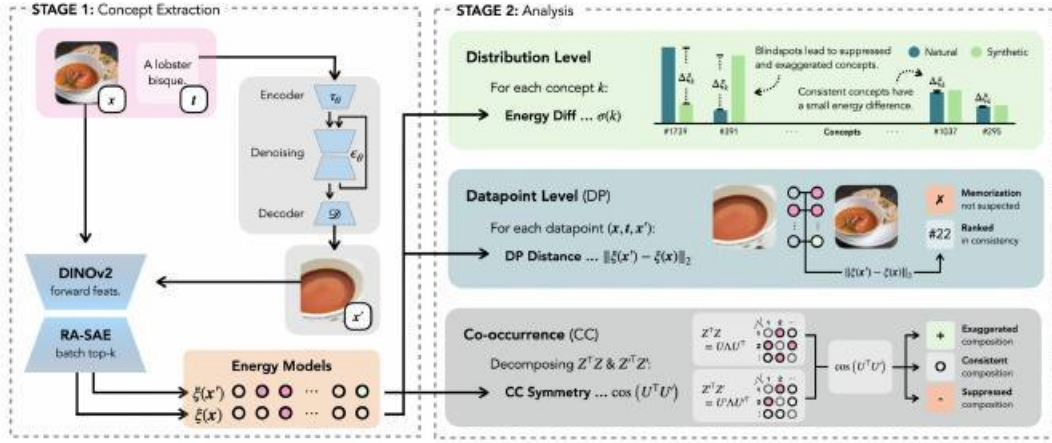


Figure 3: **Concept Extraction Pipeline.** For a triplet  $(x, t, x')$ , each image is embedded with DINOv2 and processed by a RA-SAE into sparse concept embeddings, yielding energy models  $\xi(x)$  and  $\xi(x')$  respectively. In particular, directions  $\mathbf{a}_k$  (reflected by the independent constraint in Def 1) then the SAE can separate those concepts into individual latent dimensions [136]. The activation assigned by the SAE to a dimension therefore functions as an empirical approximation of the ground-truth energy function for the concept modeled by that dimension.

Formally, let  $f : \mathcal{X} \rightarrow \mathbb{R}^d$  denote the DINOv2 feature extractor. For each natural or synthetic image  $x$  in  $\mathcal{D}_{\mathcal{X}}$  or  $\mathcal{D}'_{\mathcal{X}}$ , we compute  $\mathbf{a} = f(x) \in \mathbb{R}^d$ . For  $n$  image-text pairs, the natural and generated features are stacked into matrices  $\mathbf{A}, \mathbf{A}' \in \mathbb{R}^{n \times d}$ . Each feature vector is then decomposed by an SAE into a sparse combination of learned concept atoms. Let  $\mathbf{D} \in \mathbb{R}^{d \times K'}$  be a dictionary of  $K'$  concept vectors, and let  $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$  be the encoder that maps features to sparse concept codes. Applying  $\Psi$  row-wise produces the activation matrix  $\mathbf{Z} = \Psi(\mathbf{A}) \in \mathbb{R}^{n \times K'}$ , with each row  $\mathbf{z}_i = \Psi(\mathbf{a}_i)$  representing the concept decomposition of one image. The decoder reconstructs features as  $\mathbf{Z}\mathbf{D}^T$ , and the SAE objective minimizes reconstruction error subject to sparsity and non-negativity:

$$\min_{\Psi, \mathbf{D}} \|\mathbf{A} - \Psi(\mathbf{A})\mathbf{D}^T\|_F^2 \quad \text{s.t.} \quad \Psi(\mathbf{A}) \geq 0, \|\Psi(\mathbf{A})_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Standard SAEs can converge to arbitrarily oriented dictionaries, making downstream concept analyses sensitive to random initialization. To obtain a reproducible and data-grounded concept basis, we use the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary  $\mathbf{D}$  to be a convex combination of training examples, thereby tying each learned atom to the empirical support rather than to an arbitrary rotated basis. Specifically,  $\mathbf{D} = \mathbf{W}\mathbf{A}$  with  $\mathbf{W} \in \Omega_{K', n}$ , where  $\Omega_{K', n}$  is the set of row-stochastic matrices in  $\mathbb{R}^{K' \times n}$ :

$$\Omega_{K', n} := \left\{ \mathbf{W} \in \mathbb{R}^{K' \times n} \mid \mathbf{W} \geq 0, \mathbf{W}\mathbf{1} = \mathbf{1} \right\}. \quad (3)$$

Consequently, every atom  $\mathbf{D}_i$  lies in the convex hull of the data  $\text{conv}(\mathbf{A})$ , and every reconstruction  $\mathbf{Z}\mathbf{D}^T$  lies in the conic hull of the data  $\text{cone}(\mathbf{A})$ . This constraint anchors learned concepts to the support of the data distribution [56]. After training, the SAE yields aligned sparse codes  $\mathbf{Z}$  for natural images and  $\mathbf{Z}'$  for their generated counterparts. *These codes capture the same prompt-conditioned visual semantics in terms of shared, interpretable concepts, with the activation value of the concept serving as energy values for our analysis of conceptual blindspots.* Thus, for each prompt and associated natural image  $(t, x)$ , the pipeline synthesizes  $x'$  and maps both images into a shared sparse concept space, producing  $(z, z')$  for direct comparison of natural and generated visual content.

This pipeline operationalizes conceptual fidelity: a captioned image  $(t, x)$  and its generated counterpart  $x'$  are encoded by DINOv2 and RA-SAE into interpretable concept representations  $(z, z')$ , which provide the empirical basis for measuring concept shifts induced by the generative process.

## 5 Results

We evaluate four generative image models trained on LAION-5B or related derivatives—SD 1.5, SD 2.1, PixArt, and Kandinsky—using  $|\mathcal{X}| = 10,000$  image-text pairs

and their prompt-matched generations (Fig. 4). The analysis is organized at three complementary levels (Fig. 3): a **distribution-level** evaluation identifies suppressed and exaggerated concepts; a **datapoint-level** analysis localizes failures associated with ambiguity, omission, and memorization; and a **compositional** analysis characterizes distortions in concept co-occurrence geometry.

The empirical study is supported by the interactive exploratory tool shown in Fig. 5. The web interface centers on UMAP projections of concept representations, enabling direct visualization and comparison of concept-level energy differences. It is publicly available at a software repository, together with pre-computed energy-difference data for the four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). The analyses below use this interface to connect global distributional patterns with inspectable individual examples. Its primary functionalities are:

- **Contrast different models and architectures.** For each evaluated model, the tool provides a UMAP visualization over all 32,000 RA-SAE concepts. Each point denotes one concept and is color-coded by its energy difference.
- **Inspect concepts.** Each concept card reports key statistics, representative real and generated images ( $x$ ,  $x'$ ), and visualized co-occurrence patterns.
- **Explore blindspots.** Beyond UMAP and per-concept views, the tool provides global rankings of suppressed and exaggerated blindspots, making the most pronounced concept-level deviations directly inspectable.

Rather than exhaustively analyzing a single failure mode, we use the tool to organize high-level findings that demonstrate how the framework supports customizable concept-level investigation across models, concepts, and individual generations.



Figure 4: **Representative image-caption pairs** from LAION-5B and prompt-matched generations from SD 1.5, SD 2.1, PixArt, and Kandinsky.

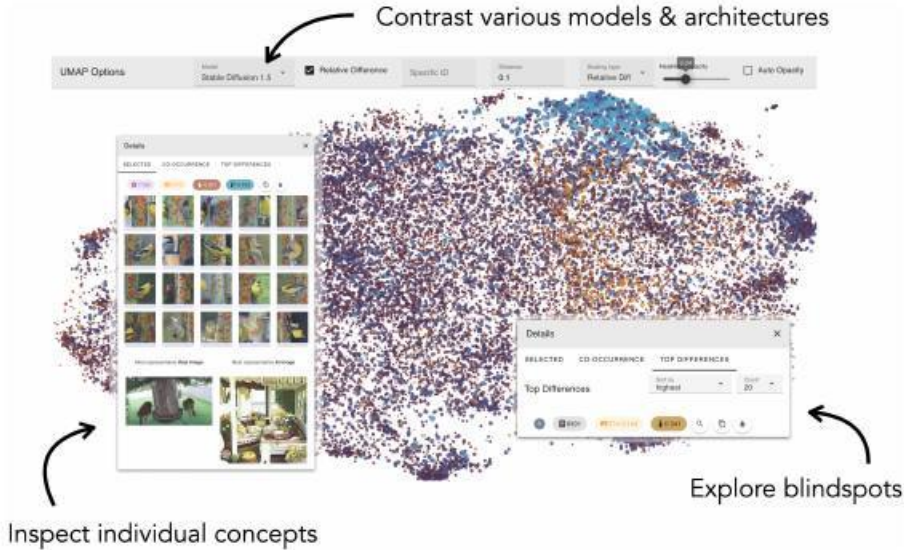


Figure 5: **Overview of the Exploratory Tool.** The web interface presents a UMAP projection for each evaluated model, with each dot corresponding to one concept and color indicating its energy difference. Selecting a concept opens a detail panel containing illustrative images, summary statistics, representative natural and generated images  $x$  and  $x'$ , an ordered list of co-occurrences, and global rankings of blindspots.



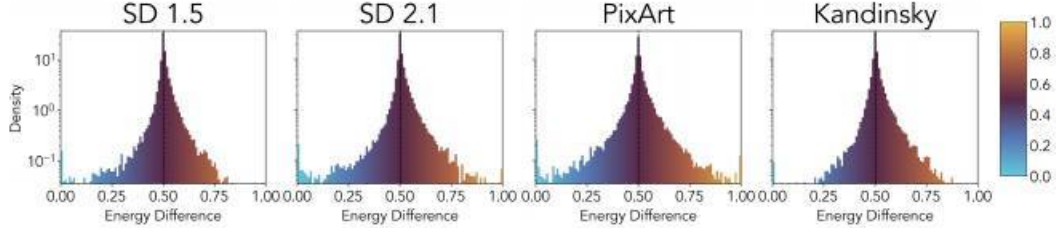


Figure 6: **Concept Energy Distribution.** Log-scale histograms of energy differences  $\delta(k)$  over 32,000 concepts for each evaluated model, comparing concept prevalence in natural and synthesized images. Values left of zero indicate suppressed concepts; values right of zero indicate exaggerated concepts.

## 5.1 Generative Image Models Cannot Produce Unseen Concepts

To measure concept-level disparities between generated and natural images, we first analyze the marginal energy difference  $\delta(k)$  for the 32,000 RA-SAE concepts. As defined in Sec. 4  $\delta(k)$  measures the relative prevalence of concept  $k$  in synthesized images compared with natural images. Values  $\delta(k) < 0.1$  indicate under-representation, or suppression, whereas values  $\delta(k) > 0.9$  indicate over-representation, or exaggeration. Fig. 6 reports the resulting distributions for the four evaluated models. All four models exhibit heavy-tailed histograms with substantial mass near both extremes, demonstrating that concept-coverage errors are not confined to isolated cases. The left tail, corresponding to suppressed concepts, is consistently denser and longer than the right tail, indicating a stronger tendency toward omission than exaggeration. This asymmetry is quantified by negative skewness in every model: Skewness =  $-0.54$  for SD 2.1,  $-0.40$  for both SD 1.5 and PixArt, and  $-0.23$  for Kandinsky.

Although all four models contain both suppressed and exaggerated concepts, their profiles are not identical. PixArt, for example, exhibits a wider spread, consistent with a more dispersed suppressed-concept distribution. The shared left-skew establishes concept omission as a recurring property across the evaluated models, while the model-specific distributional shapes motivate the structural analysis in the following sections.

## 5.2 Structure and Specificity of Conceptual Blindspots

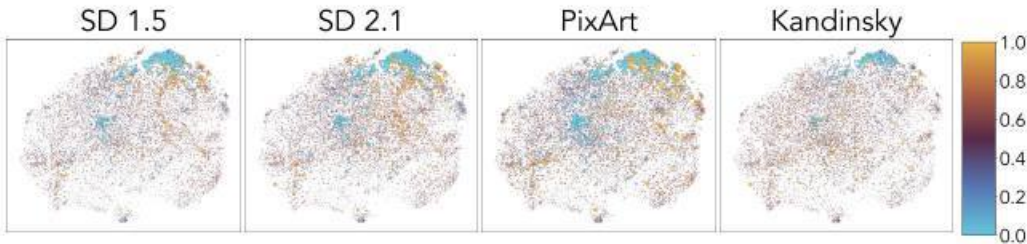


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts, colored by energy difference  $\delta(k)$  between natural and synthesized images. Clustered regions reveal structured conceptual blindspots, with suppressed concepts shown in blue and exaggerated concepts in yellow.

Having established marginal concept-frequency discrepancies, we next examine whether blindspots possess coherent organization in concept space. We embed all 32,000 concepts into two dimensions with UMAP applied to the sparse codes and color each concept by  $\delta(\cdot)$ . Fig. 7 shows that blindspots form distinct clusters across all models. These clusters often correspond to contiguous regions of suppressed concepts, indicating that blindspots are structured rather than randomly distributed and may reflect shared properties of the training data or architectural priors. We then quantify the cross-model consistency of these structures by comparing concept-level  $\delta(\cdot)$  vectors. Fig. 8 reports scatter plots and pairwise Pearson correlations between SD 1.5 and the other models. SD 1.5 and SD 2.1 are strongly correlated ( $r = 0.82$ ), consistent with their related architectures and training pipelines.



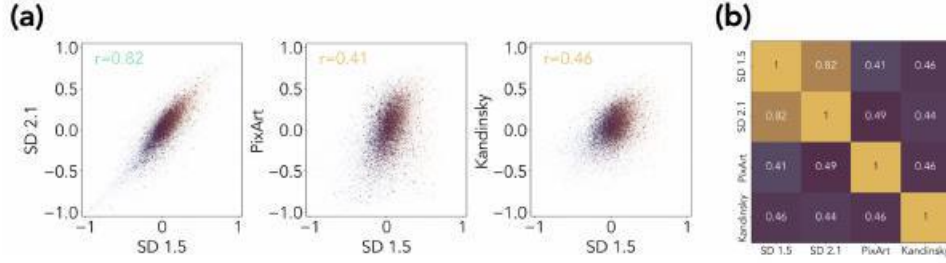


Figure 8: **Cross-Model Concept Energy Correlation.** Pairwise scatter plots of  $\delta(k)$  across all four evaluated models, with Pearson correlation coefficients reported in the upper left. Strong alignment between SD 1.5 and SD 2.1 contrasts with weaker correlations involving other architectures, indicating model-specific blindspot profiles. **Correlation Matrix of Conceptual Blindspots.** Heatmap of pairwise Pearson correlation coefficients for  $\delta$  between all models, quantifying how much conceptual blindspot structure is shared across models.

Correlations with PixArt and Kandinsky are lower— $r = 0.41$  and  $r = 0.46$ , respectively—showing that different model families concentrate their blindspots in different regions of the concept space.

Together, these results establish that conceptual blindspots contain both shared and model-specific components. Shared blindspots plausibly reflect properties of the data distribution, whereas model-specific blindspots point to differences in architecture, training dynamics, or capacity; the qualitative analysis below therefore examines both universal and unique cases.

### 5.3 Qualitative Examples of Blindspots

We next visualize specific suppressed and exaggerated blindspots to clarify the kinds of concepts captured by each regime. Specifically, Fig. 9a shows a conceptual blindspot suppressed by all evaluated models: each model fails to reproduce the concept **solid white on documents**. Despite captions that explicitly reference this concept, none of the generated images reflects the intended visual semantics, suggesting that this region of the concept space is systematically under-sampled across models. Fig. 9b, by contrast, highlights a model-specific blindspot: the concept **pan** is accurately captured by three models but conspicuously absent from generations produced by Kandinsky. This qualitative contrast reinforces the finding from Sec. 5.2 that cross-model agreement is high for some blindspots but limited for others.

Conversely, Fig. 10 presents an exaggerated blindspot in which the concept **shadow under animal** is over-emphasized in generated images. Although shadows are visually plausible in these scenes, their consistent and pronounced rendering across models, relative to the more nuanced and variable occurrences in natural images, indicates an overactive prior. Despite attempts to identify concepts uniquely exaggerated by a specific model, we did not find clear examples, suggesting that the observed exaggerations are approximately universal within the evaluated set.

Overall, these examples show how conceptual blindspots manifest in generated outputs and



Figure 9: **Examples of Suppressed Conceptual Blindspots.** The natural images and synthesized images  $c_k$  and  $t$ , shown alongside four synthesized images  $x'$  generated using  $S$ . The universal blindspot is present in all evaluated models; the unique blindspot is only present in Kandinsky.

Figure 10: **Example of an Exaggerated Conceptual Blindspot.** The natural images and synthesized images  $c_k$  and  $t$ , shown alongside four synthesized images  $x'$  generated using  $S$ . The universal blindspot is present in all evaluated models; the unique blindspot is only present in Kandinsky.

demonstrate that the energy-based diagnostic surfaces both shared and model-specific failure modes. The consistent failure to reproduce clear or solid background elements, such as whitespaces on documents, is especially informative because it suggests that some failure patterns may arise from architectural constraints or training-data biases that transcend individual model idiosyncrasies.

While these aggregate-level analyses are informative, they motivate a finer-grained question: do blindspots appear only as population-level discrepancies, or do they also manifest at individual datapoints? This perspective probes blindspot mechanisms more directly by exposing cases of prompt misinterpretation, latent memorization, or both.

#### 5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we examine individual natural vs. generated image pairs for which the  $\delta(\cdot)$  values averaged across all concepts exhibit the largest and smallest differences. This analysis makes model successes and failures interpretable at the example level; the latter often arise from prompt ambiguity or memorization artifacts. For example, Fig. 11a shows instances with near-zero difference in average  $\delta(\cdot)$  values. In these cases, the generated images are conceptually nearly indistinguishable from the originals. Qualitative inspection indicates that this agreement does not necessarily reflect faithful prompt semantics; instead, it can arise from replication of memorized templates, including repetitive visual structures such as clothing outlines or object arrangements that may correspond to frequent training patterns. By contrast, Fig. 11b illustrates samples that are among the largest  $\delta(\cdot)$  values. These samples indicate substantial conceptual divergence between the synthesized and natural image. Although some discrepancies can be attributed to underspecified or noisy captions, others reveal genuine blindspots: the prompt describes a clear concept faithfully present in  $\mathcal{D}_X$ , yet the model fails to realize it in  $\mathcal{D}'_X$ . This failure suggests that, even when language grounding is adequate, certain concepts remain outside the model’s effective generative repertoire.



Figure 11: • **Datapoint-level Conceptual Alignment.** (a) Examples with minimal energy differences where models appear to memorize training patterns. (b) Examples with large differences where significant concept divergences due to prompt ambiguity or model limitations occur.



## 383 5.5 Analyzing Post-Training Effects

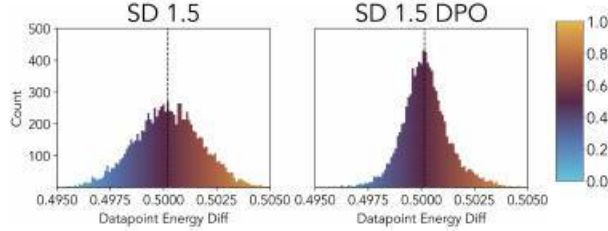


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between the synthesized and natural distribution of SD 1.5 models with and without DPO.

384 Post-training protocols, including safety fine-tuning, have been argued to reduce the diversity of  
 385 model generations [140]. Given the pipeline’s ability to isolate concept-level differences between  
 386 model generations and the ground-truth DGP, we next use it to characterize the effects of DPO—a  
 387 popular safety fine-tuning protocol [141]. Specifically, we compare two variants of SD 1.5: one  
 388 trained with DPO and one without DPO. For each image pair  $(\mathcal{D}_X, \mathcal{D}'_X)$ , we compute the  $\ell_2$  norm of  
 389 the difference between their internal concept energy vectors,  $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$ . Fig. 12 presents  
 390 a histogram of these datapoint-wise energy differences. The DPO-enhanced model exhibits both a  
 391 lower median and a narrower spread, indicating a more consistent distribution of generated concepts  
 392 relative to the ground-truth DGP. This pattern suggests that DPO may regularize the model’s concept  
 393 distribution, encouraging outputs that more closely reflect the semantic content of the observed inputs.  
 394 While our analysis does not disentangle the specific inductive biases introduced by DPO, these results  
 395 provide empirical evidence that its optimization objective, which favors human-preferred generations,  
 396 indirectly promotes closer alignment with the training distribution. In particular, it reduces *both over*  
 397 *and under* activation of individual concepts relative to the natural baseline. These findings highlight  
 398 the utility of the pipeline for characterizing downstream post-training effects not only in terms of  
 399 output quality, but also in terms of how they reshape the conceptual geometry of the model’s output  
 400 space.

## 401 5.6 Conceptual Misalignment as a Function of Empirical Frequency

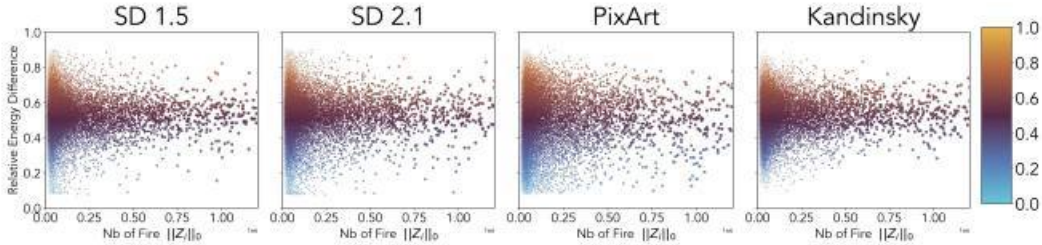


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots show the relationship between concept frequency (x-axis) and energy difference (y-axis) across the four evaluated models. Each point represents one concept, with marker size proportional to its activation frequency.

402 We next test whether conceptual blindspots are not merely architectural artifacts, but also emerge  
 403 from distributional peculiarities of particular features. We examine whether concepts that are rarely  
 404 activated in natural images—the long tail of the data distribution—are more likely to become  
 405 blindspots in generative models. Specifically, we process the natural dataset  $\mathcal{D}_X$  through the trained  
 406 SAE and compute, for each concept  $k$ , its empirical frequency  $\|\mathbf{Z}_{:,k}\|_0$ , where  $\mathbf{Z}_{:,k}$  denotes the  
 407 activations of concept  $k$  across all images. We then correlate this frequency with the absolute energy  
 408 difference observed across generated outputs. Fig. 13 visualizes this relationship for all evaluated  
 409 models. Concepts with higher frequency in natural data tend to show lower energy discrepancies,  
 410 whereas rare concepts—especially suppressed ones ( $\delta(k) < 0.5$ )—exhibit substantial alignment  
 411 errors. This pattern indicates that many blindspots arise not from randomness or isolated model  
 412 quirks, but from systematic effects tied to long-tail concept distributions. Addressing these issues may  
 413 therefore require not only architectural changes, but also data-centric strategies such as reweighting  
 414 or augmentation.

## 6 Discussion

Our analysis reveals multiple conceptual blindspots in four popular generative image models. The results presented here, however, only scratch the surface: each individual finding could warrant its own dedicated investigation. Rather than delving deeply into any one of these questions, we instead showcase the versatility of our method and exploratory tool. Out of the box, they allow for a systematic identification of concepts that models struggle to generate, detection of memorization artifacts, discovery of datapoints with insufficient captions, quantification of post-training effects, and characterization of conceptual shifts across model architectures. We thus open space for follow-up work to extend the depth of analysis, scope of evaluated architectures, and inquiry into root causes of conceptual blindspots. Future work could also explore hierarchical representations of concepts in RA-SAE to allow for a more nuanced analysis.

**Limitations.** We wish to highlight several limitations of our work. By relying on DINOv2 and RA-SAE for concept extraction and representation, our approach is inherently constrained to the kinds of concepts these models capture; concepts poorly represented by them will escape our analysis. Additionally, while our sample size of 10,000 images is substantial, it may not fully capture the long tail of rare concepts, concept co-occurrence, or other compositional statistics.



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# Appendix

835

836

837

## Table of Contents

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838

|                             |           |
|-----------------------------|-----------|
| <b>A Experimental Setup</b> | <b>23</b> |
|-----------------------------|-----------|

839

|                       |    |
|-----------------------|----|
| A.1 Observation Space | 23 |
|-----------------------|----|

840

|                        |    |
|------------------------|----|
| A.2 Synthesized Images | 23 |
|------------------------|----|

841

|                                                                                                                   |    |
|-------------------------------------------------------------------------------------------------------------------|----|
| A.3  Distribution Level Analysis | 23 |
|-------------------------------------------------------------------------------------------------------------------|----|

842

|                                                                                                                |    |
|----------------------------------------------------------------------------------------------------------------|----|
| A.4  Datapoint Level Analysis | 24 |
|----------------------------------------------------------------------------------------------------------------|----|

843

|                                                                                                              |    |
|--------------------------------------------------------------------------------------------------------------|----|
| A.5  Co-occurrence Analysis | 24 |
|--------------------------------------------------------------------------------------------------------------|----|

844

|                                         |           |
|-----------------------------------------|-----------|
| <b>B Custom RA-SAE Training Details</b> | <b>25</b> |
|-----------------------------------------|-----------|

845

|                                  |           |
|----------------------------------|-----------|
| <b>C Computational Resources</b> | <b>26</b> |
|----------------------------------|-----------|

846

|                               |           |
|-------------------------------|-----------|
| <b>D Autointerpretability</b> | <b>27</b> |
|-------------------------------|-----------|

847

|                                                        |           |
|--------------------------------------------------------|-----------|
| <b>E Additional Results: Model-Specific Blindspots</b> | <b>29</b> |
|--------------------------------------------------------|-----------|

848

|                                                                                     |           |
|-------------------------------------------------------------------------------------|-----------|
| <b>F Additional Results: Higher-order Blindspots with Compositional Discrepancy</b> | <b>31</b> |
|-------------------------------------------------------------------------------------|-----------|

849

|                                                       |           |
|-------------------------------------------------------|-----------|
| <b>G Additional Results: Blindspot Stress Testing</b> | <b>33</b> |
|-------------------------------------------------------|-----------|

850

|                                        |    |
|----------------------------------------|----|
| G.1 Bird Feeder Blindspot in Kandinsky | 33 |
|----------------------------------------|----|

851

|                                         |    |
|-----------------------------------------|----|
| G.2 Glossy DVD Disc Blindspot in SD 1.5 | 34 |
|-----------------------------------------|----|

852

|                                                  |           |
|--------------------------------------------------|-----------|
| <b>H Additional Examples: Synthesized Images</b> | <b>35</b> |
|--------------------------------------------------|-----------|

853

|                                                       |           |
|-------------------------------------------------------|-----------|
| <b>I Concentration bounds for <math>\delta</math></b> | <b>36</b> |
|-------------------------------------------------------|-----------|

854

|                                                                                  |           |
|----------------------------------------------------------------------------------|-----------|
| <b>J Monotonicity and Calibration-Free Interpretation of <math>\delta</math></b> | <b>37</b> |
|----------------------------------------------------------------------------------|-----------|

855

|                                                |           |
|------------------------------------------------|-----------|
| <b>K Stability of FID Under SAE Embeddings</b> | <b>38</b> |
|------------------------------------------------|-----------|

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856

857

858



## A Experimental Setup

This section details the experimental setup for our analysis of four popular generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or its subsets/derivatives. The code is available at a software repository.

### A.1 Observation Space

The observation space is constructed by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our domain of natural images. Due to concerns with CSAM and other unsafe content in the dataset, the original data release is no longer available. A substitute release of a subset of this dataset with additional filtering of the unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe> is used.

The sampling procedure consists of: (1) loading the full LAION dataset using the Hugging Face `datasets` library, (2) performing validation to ensure proper URL structure and resource availability via HTTP HEAD requests, and (3) employing random sampling with replacement until reaching the target count of 10,000 valid samples. This approach, yielding  $D_G$  with  $(x, t)$  tuples, ensures our observation space contains accessible image-text pairs for comparative analysis of a dataset of image URLs whose large portion has been made unavailable since original release. Additional examples of synthesized images are shown in Appendix H.

### A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset  $D_{g_\theta}$  to have a one-to-one correspondence with  $D_G$ , yielding triplets  $(x, x', t)$ . Specifically, given the 10,000 image-text pairs  $(x, t)$  from  $D_G$ , we use  $t$  to synthesize counterpart images  $x'$  using each generative model  $g_\theta$ .

The synthesis process follows the standard text-to-image generation pipeline for each model architecture, implemented using the Hugging Face `diffusers` library, where the models are loaded at mixed precision (fp16). All synthetic images are generated at  $512 \times 512$  pixel resolution with default parameters.

**Stable Diffusion 1.5.** The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

**Stable Diffusion 1.5 + DPO.** The DPO variant of SD 1.5 (used in the analysis in Sec. 5.5) follows the baseline SD 1.5 implementation, but replaces the UNet component with a DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>.

**Stable Diffusion 2.1.** The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

**Kandinsky.** The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

**PixArt.** The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

### A.3 Distribution Level Analysis

**Section 5.1.** We compute energy differences  $\delta(\cdot)$  across all 32,000 concepts for each evaluated model. The sigmoid transformation with temperature  $T = 0.8$  is applied during normalization. The resulting values are visualized as log-scale density histograms with 100 bins spanning  $[0, 1]$ .

904 **Section 5.2** We embed the complete set of 32,000 concepts into two-dimensional space using  
 905 UMAP applied to the sparse concept codes. Each point in this UMAP represents an individual con-  
 906 cept, colored according to its energy difference  $\delta(\cdot)$ , emphasizing both suppressed and exaggerated  
 907 blindspots. To quantify cross-model consistency, we compute pairwise Pearson correlation coeffi-  
 908 cients between  $\delta(\cdot)$  vectors of all model pairs, producing both scatter plots and correlation matrices.  
 909 This analysis reveals whether blindspots cluster in conceptual space and identifies model-specific  
 910 versus universal patterns of conceptual blindspots.

911 **Section 5.3** We rank all 32,000 concepts by their energy difference  $\delta(\cdot)$ , and manually examine the  
 912 extrema (both suppressed and exaggerated blindspots). For suppressed blindspots, we select concepts  
 913 with  $\delta(\cdot) < 0.1$ ; for exaggerated blindspots, we choose those with  $\delta(\cdot) < 0.9$ . Presented examples  
 914 are manually annotated with textual descriptions of the respective concepts through inspection of  
 915 their most activating images and spatial attention patterns. We outline ongoing efforts to automate  
 916 this concept interpretation in Appendix D

917 **Section 5.5** We compare 1.5 with and without DPO in the following fashion: for each image  
 918 pair  $(x, x')$ , we compute the L2 norm of the difference between their concept energy vectors  
 919  $\|\xi(x') - \xi(x)\|_2$ . We apply a sigmoid transformation with temperature  $T = 0.8$  to the element-wise  
 920 differences before taking their mean. This yields datapoint-wise energy differences that quantify how  
 921 much each generated image deviates from its natural counterpart in concept space. Finally, these  
 922 differences are visualized as overlapping histograms, contrasting both model variants.

923 **Section 5.6** For each concept  $c_k$ , its empirical frequency  $\|\mathbf{Z}_{:,i}\|_0$  (the count of non-zero activations  
 924 across the natural dataset) is counted. A sigmoid normalization with temperature  $T = 0.4$  is then  
 925 applied to the energy differences  $\delta(\cdot)$ . The analysis is visualized using scatter plots where the x-axis is  
 926 the empirical concept frequency and the y-axis is the sigmoid-transformed energy difference. The  
 927 point sizes are proportional to activation frequency and point colors are proportional to the magnitude  
 928 of energy differences.

#### 929 A.4 • Datapoint Level Analysis

930 **Section 5.4** For each image pair  $(x, x')$ , we compute the L2 norm of the difference between their  
 931 concept energy vectors  $\|\xi(x') - \xi(x)\|_2$ . This yields a scalar measure of conceptual divergence for  
 932 each image pair. The samples are ranked by their energy differences. Minimal divergence indicate  
 933 potential memorization artifacts and maximal divergence point to significant conceptual failures. This  
 934 analysis enables qualitative inspection of specific failure modes.

#### 935 A.5 • Co-occurrence Analysis

936 **Appendix F** For both the natural and synthesized data  $D_G$  and  $D_{g\theta}$ , concept co-occurrence  
 937 patterns are analyzed through the co-activation matrix  $Z^T Z$ , which holds pair-wise correlations  
 938 in concept usage. Spectral analysis is performed to examine the dominant conceptual directions  
 939 using eigendecomposition. The alignment between natural and synthetic co-occurrence structures is  
 940 assessed using cosine similarity heatmaps between the top-100 eigenvectors of each co-occurrence  
 941 matrix. These  $100 \times 100$  similarity matrices are visualized as square heatmaps where perfect diagonal  
 942 alignment would indicate identical principal concept axes, while off-diagonal patterns reveal would  
 943 revolve rotations and mismatches in compositional geometry.



## 944 B Custom RA-SAE Training Details

945 This section summarizes the training details of our custom archetypal SAE (RA-SAE) [56]. The  
946 model is open-sourced at a software repository.

947 **Dataset.** The auto-encoder is trained on the complete ImageNet-1k training split, ( $\approx 1.28$  M)  
948 RGB images. Each image is converted to 261 visual tokens using DINOv2 [135]; tokens are fed  
949 to the SAE without class or position embeddings. The total number of training tokens is therefore  
950  $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$ .

951 **Dictionary.** The dictionary has 32,000 concept dimensions. For the sparse activation rule, top- $k$   
952 masking with  $k = 5$ , is used; activations outside the largest five per input are set to 0. The weights  
953 are initialized using Xavier/Glorot. The training is conducted at mixed precision (fp16), with the last  
954 ten epochs performed at full precision.

955 **Optimizer and Schedule.** The model is trained for 50 epochs using base AdamW ( $\beta_1 = 0.9$ ,  
956  $\beta_2 = 0.999$ ) optimizer is employed with weight decay set to  $10^{-5}$ . Linear warm-up is applied on the  
957 first 5% of steps, followed by cosine decay from  $\eta_{\max} = 5 \times 10^{-4}$  to  $\eta_{\text{final}} = 10^{-6}$ . MSE loss is used  
958 alongside an auxiliary term penalizing activations that never enter the top- $k$  set, where  $\lambda = 10^{-5}$ .

## 959 C Computational Resources

960 This section summarizes the GPU resources used for training and experiments in support of this  
961 paper. In total, we used approximately 202 GPU-hours on NVIDIA H100s and H200s.

962 **RA-SAE.** Trained for approximately 24 GPU-hours on three NVIDIA H100s.

963 **Synthesized Images.** Generating the full  $D_{g\theta}$  (see Appendix A) took roughly 5 hours per generator  
964 when distributed across four NVIDIA H200 GPUs. With five generators, this totaled approximately  
965 100 GPU-hours on a NVIDIA H200.

966 **● Distribution-Level Analysis.** Extracting energy differences at the distribution level took about 3  
967 hours per generator on a single NVIDIA H200 GPU (total  $\sim 15$  GPU-hours).

968 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per  
969 datapoint, and ranking datapoints per concept also took approximately 3 hours per generator on one  
970 H200 GPU (total  $\sim 15$  GPU-hours).



## D Autointerpretability

In an effort to automate the interpretation of concepts identified by the RA-SAE, we qualitatively evaluated the ability of Vision-Language Models (VLMs) to describe them. In particular, we queried ChatGPT 4o via the OpenAI API using various prompts and formats of an image  $x''$  in which the given concept was salient, recording the description generated by the VLM. In this section, we highlight the findings from our exploration. For evaluation purposes, we allowed the VLM to provide its full reasoning; to make this a scalable solution, a simple modification of the prompt—asking the VLM to provide only the description—would filter out the reasoning.

First, we provided the VLM with two images: a raw version of  $x''$  and  $x''$  with a heatmap superimposed to localize the intensity of the concept:

**Input**



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DINOv2-based SAE. Describe what is this concept.

**Output**

The concept identified by the DINOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

**Reasoning:**

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

As seen in the example, the VLM did not interpret the heatmap visualization correctly. Instead of describing the wood texture in the background—the actual salient region—it described the ring in the foreground, which received no attention from the RA-SAE.

To determine whether this was a prompting issue, we tested the same scenario again—this time using an expanded prompt that explained the heatmap in greater detail:

**Input**



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

**Output**

The concept identified by the DINOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

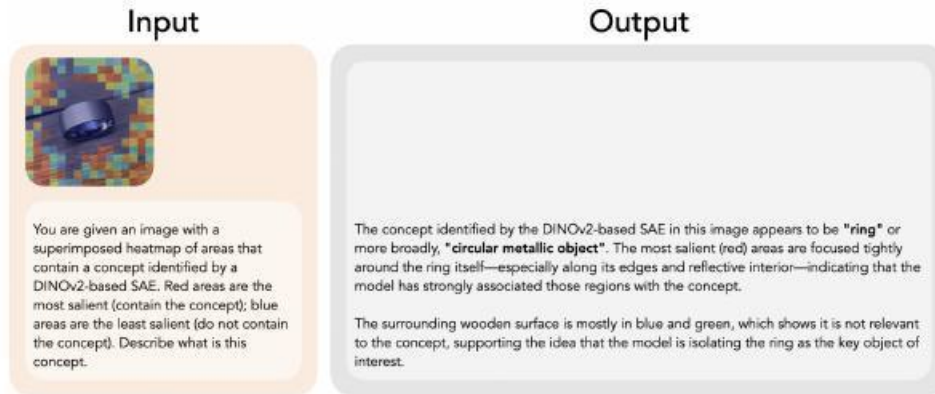
Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

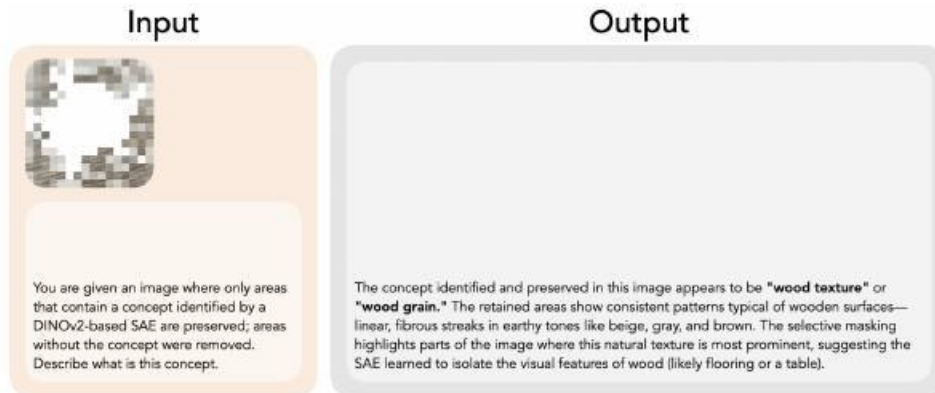
This prompt modification did not resolve the issue, as the VLM still identified the ring as the prominent concept in the image. Several other prompting techniques, including those describing the heatmap in even greater depth (e.g., by including the code used for its generation), were explored with similar results.

990 We next tested providing the VLM only with the heatmap-superimposed  $x''$  and a similar prompt  
991 explaining how to interpret the heatmap:



992 Once again, the result failed to identify the wooden background as the prominent concept in the image  
993 and instead focused on the ring. As in the previous case, we experimented with various prompting  
994 strategies using this image format, but the results remained largely unchanged.

995 From these observations, we concluded that the heatmap alone is insufficient to guide the VLM's  
996 attention to the salient areas. The presence of less relevant regions appears to distract the model. To  
997 address this, we replaced the heatmap-superimposed  $x''$  with an alpha-masked version of  $x''$ :



998 As shown in the example, the VLM was now able to correctly identify wood texture as the dominant  
999 concept. We observed consistent success across a range of concepts and images.

1000 The formatting of  $x''$  and the prompt shown above yielded the most reliable results in our qualitative  
1001 evaluation. However, we note that this evaluation is limited by its qualitative nature (due to the  
1002 absence of ground truth annotations) and its focus on a single VLM. We hope future work on the  
1003 autointerpretability of SAE concepts can build on and expand this analysis.



## 1004 E Additional Results: Model-Specific Blindspots



Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

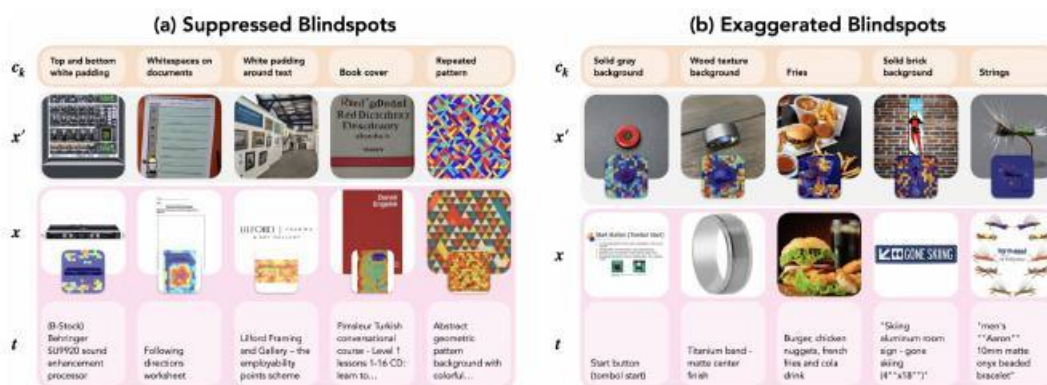


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

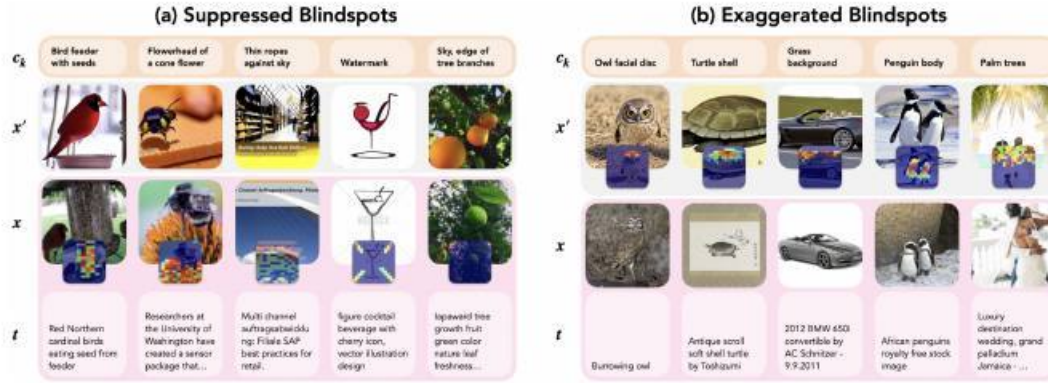


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.



## F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, our analysis has centered on individual concept activations. Yet visual scenes are rarely composed of isolated concepts; instead, they are structured through rich and structured co-occurrence patterns that encode compositional semantics. We now examine whether generative models capture this higher-order structure by analyzing the co-activation matrix  $Z^T Z$ , which reflects pairwise correlations in concept usage.

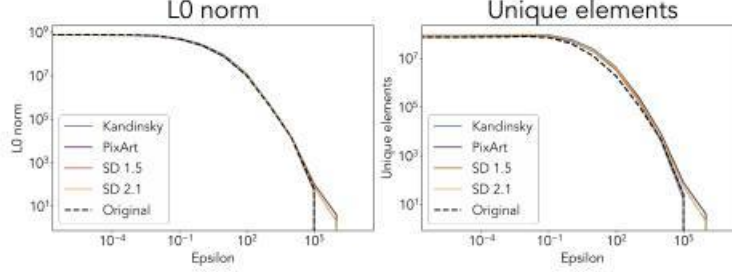


Figure 18: **Sparsity and Structural Divergence.** On the left: L0 norm of the co-occurrence matrix  $Z Z^T$  as a function of  $\epsilon$  (threshold), indicating how many entries remain active in each model. On the right: Number of unique entries in the synthesized distribution relative to the natural distribution. All evaluated models preserve global sparsity structure, but diverge in activation content.

Surprisingly, when assessed at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As shown in Fig. 18 (left), the  $\ell_0$  norm of  $Z^T Z$  – thresholded at varying  $\epsilon$  values – tracks closely between the natural and synthesized distributions across all models. This indicates that the gross connectivity of the conceptual graph, i.e., which concepts tend to co-activate at all, is well preserved. Formally, one can deem  $Z^T Z$  as the adjacency matrix of a weighted, undirected graph over concepts, where edge weights reflect co-activation strength across the dataset.

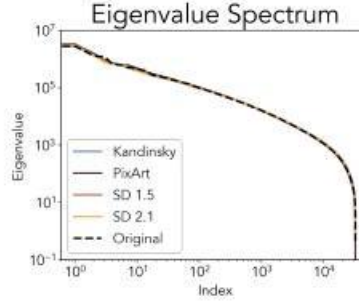


Figure 19: **Spectral Structure of Co-occurrence.** Log-log plot of the eigenvalue spectra from co-occurrence matrices  $Z Z^T$  across models and the natural distribution. All evaluated models match the heavy-tailed decay of the natural distribution.

However, as illustrated in Fig. 18 (right), the specific content of these co-activations diverges: a substantial portion of entries in the model-generated  $Z^T Z$  are not shared with the natural baseline. This suggests that while the capacity for compositionality is retained, the identity of active pairings may shift, potentially reflecting model specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 shows the eigenvalue spectra of the co-occurrence matrices for each model and the natural distribution. All spectra exhibit a heavy-tailed decay, consistent with power-law behavior, indicating that generative models preserve the overall rank structure and variance allocation across conceptual dimensions.

Further, we examine the alignment of dominant conceptual directions via cosine similarity heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20). While all models exhibit partial diagonal alignment—implying overlap in principal concept axes—the

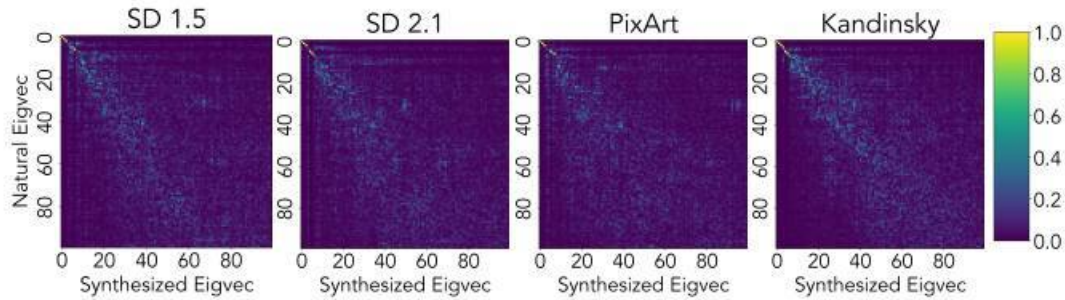


Figure 20: ● **Concept Basis Similarity.** Cosine similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices  $ZZ^T$ . Diagonal structure shows alignment of dominant conceptual directors, with varying degrees of alignment across the four models.

1031 off-diagonal entries reveal rotations and mismatches in higher modes, reflecting evident deviations in  
 1032 compositional geometry.

1033 Together, these findings reveal that diffusion models approximate the global shape of concept co-  
 1034 activation surprisingly well, yet deviate in subtle and structured ways when examined through  
 1035 the spectral lens. Such higher-order discrepancies may underpin failures in generating coherent,  
 1036 multi-object scenes or relational concepts.



## 1037 G Additional Results: Blindspot Stress Testing

1038 To stress-test the blindspots identified by our method, we gathered a range of prompts describing  
 1039 these blindspots and used them to generate many images. We then contrasted the outputs from models  
 1040 in which the concept was identified as a blindspot with those in which it was not.

1041 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*  
 1042 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*  
 1043 *text-to-image model. Return these as a list of strings in Python.* Five images were generated per  
 1044 prompt and analyzed using our custom RA-SAE model (see Appendix B), which ranked them by  
 1045 the intensity with which the desired concept appeared. All images were then manually reviewed to  
 1046 determine whether the blindspot was successfully depicted.

1047 As seen in the following examples of suppressed concepts, while some aspects of the target concept  
 1048 occasionally appeared (e.g., a holder or string for the bird feeder blindspot and a round hole for  
 1049 the glossy DVD disc blindspot), the models generally failed to generate the full concept. This  
 1050 aligns with our method’s assessment and supports the validity of the stress test.

### 1051 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Examples of images generated with various prompts involving the bird feeder concept as a part of the stress testing. In Kandinsky, our method identified this concept as a suppressed conceptual blindspot, which matches the observed behavior: the model is unable to generate a corresponding image. By contrast, SD 1.5, in which this concept was not identified as a blindspot, is able to generate this concept.



Figure 22: Examples of images generated with various prompts involving the glossy DVD disc concept as a part of the stress testing. In SD 1.5, our method identified this concept as a suppressed conceptual blindspot, which matches the observed behavior: the model is unable to generate a corresponding image. By contrast, Kandinsky, in which this concept was not identified as a blindspot, is able to generate this concept.



1053 **H Additional Examples: Synthesized Images**



Figure 23: Additional image-caption pair examples from LAION-5B with matching images generated with the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

## 1054 I Concentration bounds for $\delta$

1055 In our experiments, we estimate  $\delta(k)$  using  $n = 10,000$  paired samples for each concept. While this  
 1056 budget is modest, it raises the natural question of whether it suffices to obtain reliable estimates. To  
 1057 address this, we derive a concentration bound on the empirical estimator  $\hat{\delta}_n(k)$  using McDiarmid’s  
 1058 inequality [142]. The resulting bound is tight and demonstrates that even with relatively few samples,  
 1059 we can obtain fast and accurate estimates of concept bias.

1060 **Theorem 4** (Concentration of  $\hat{\delta}_n(k)$ ). *We assume that the concept score  $\xi_k(\mathbf{x})$  takes values in  $[a, b]$   
 1061 almost surely for all images  $\mathbf{x}$  drawn from either  $\mathcal{D}_X$  or  $\mathcal{D}'_X$ . Let  $n$  paired samples  $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$  be  
 1062 drawn independently with  $\mathbf{x}_i \sim \mathcal{D}_X$  and  $\mathbf{x}'_i \sim \mathcal{D}'_X$ , and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left( \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1063 *Let  $M := b - a$  and  $L := M/4$ . Then for every  $\varepsilon > 0$ , the deviation satisfies*

$$\mathbb{P} \left( \left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left( -\frac{2n\varepsilon^2}{L^2} \right).$$

1064 *Proof.* The function  $x \mapsto \sigma(x)$  is  $1/4$ -Lipschitz, since  $|\sigma'(x)| \leq 1/4$  for all  $x$ .

1065 Viewing  $\hat{\delta}_n(k)$  as a function of the  $2n$  independent variables  $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$ , changing a  
 1066 single argument alters the inner difference of means by at most  $M/n$ , and the outer sigmoid scales this  
 1067 by at most  $1/4$ . Hence, the bounded difference constant for each coordinate is  $(M/n)(1/4) = L/n$ .

1068 By McDiarmid’s inequality [142],

$$\mathbb{P} \left( \left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left( -\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left( -\frac{2n\varepsilon^2}{L^2} \right),$$

1069 which proves the claim. □

1070 Practically, most concept scores  $\delta(k)$  are sparse, with the majority concentrated near zero and only  
 1071 a few reaching values up to 10. The concentration bound shows that even for the largest observed  
 1072 biases, a sample size of  $n = 10,000$  yields estimates of  $\hat{\delta}_n(k)$  that deviate from the true value by  
 1073 no more than a small  $\varepsilon$  with high probability. This justifies our sampling strategy and confirms that  
 1074 accurate bias measurements are attainable with limited data.



## 1075 J Monotonicity and Calibration-Free Interpretation of $\delta$

1076 Our goal when analyzing blind spots is to rank concepts by the severity of their generative bias. In  
 1077 practice, we use the score  $\delta(k)$  for this purpose. However, one may wonder whether such a score  
 1078 introduces distortions relative to more direct quantities such as the energy gap or the odds ratio.  
 1079 The following result establishes that  $\delta(k)$  is a strictly increasing reparameterization of both, and  
 1080 therefore inherits their ordering. This guarantees that no calibration is needed when using  $\delta(k)$  to  
 1081 rank concepts.

1082 **Theorem 5** (Monotonicity and Calibration of  $\delta_{g_\theta \leftrightarrow G}$ ). *For every concept index  $k$  define the energy*  
 1083 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

1084 *the associated odds ratio  $\rho_k = \exp(\Delta_k)$ , and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1085 *Then  $\delta(k)$  is a strictly increasing bijection of both  $\Delta_k$  and  $\rho_k$ , so ranking concepts by any one of*  
 1086  *$\delta(k)$ ,  $\Delta_k$ , or  $\rho_k$  produces exactly the same ordering.*

1087 *Proof.* The logistic sigmoid satisfies  $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$ ; hence  $\sigma$  and therefore  
 1088  $\delta(k) = \sigma(\Delta_k)$  grow strictly with  $\Delta_k$ . Because the exponential map is also strictly increasing and  
 1089 bijective  $\mathbb{R} \rightarrow (0, \infty)$ , setting  $\rho_k = \exp(\Delta_k)$  preserves order and gives  $\Delta_k = \log \rho_k$ . Substituting  
 1090 this identity into  $\sigma$  yields  $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$ , which is the composition of two  
 1091 strictly increasing bijections and is therefore itself strictly increasing and bijective in  $\rho_k$ . Since strict  
 1092 monotonic functions never reverse inequalities, the three quantities share the same total order over  
 1093 concepts.

1094

□

1095 Thus, ranking concepts by  $\delta(\cdot)$  is strictly equivalent to ranking them by energy gap or by conceptual  
 1096 generation odds  $\rho_k$ . No calibration is necessary, and all three quantities preserve the same total  
 1097 ordering over concepts.

## 1098 K Stability of FID Under SAE Embeddings

1099 In this section we establish a quantitative relationship between the Fréchet Inception Distance  
1100 (FID) computed in the original activation space (of dimension  $d$ ) and the FID after applying a  
1101 (potentially overcomplete) SAE dictionary  $D \in \mathbb{R}^{k \times d}$  with  $k \gg d$ . Throughout we assume that  $D$   
1102 has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

1103 We start by recalling a simple fact: if  $D$  is not overcomplete, orthogonal and  $k = d$ , then we have an  
1104 isometry between  $A$  and  $Z$ , implying that the FID is perfectly preserved. However, this case is not  
1105 realistic, we will then turn to the overcomplete case, and show we can bound FID by the extreme  
1106 singular value of  $D$ . We will work with the Wasserstein-2 metric  $\mathcal{W}_2$ , noting that FID is just  $\mathcal{W}_2^2$   
1107 specialised to Gaussians.

1108 For a probability measure  $\mu$  on  $\mathbb{R}^d$  we write  $D_\# \mu$  for its push-forward under  $D$ , i.e.  $D_\# \mu(z) =$   
1109  $\mu(D^{-1}z)$ . Denote by  $\sigma_{\min}$  and  $\sigma_{\max}$  the minimal and maximal singular values of  $D$ , equivalently  
1110 the square-roots of the extremal eigenvalues of  $DD^\top$ :

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

1111 Empirically one usually finds  $\sigma_{\min}, \sigma_{\max} \approx 1$ , but the proof does not rely on that. We will start by a  
1112 simple lemma in the case where  $D$  is not overcomplete.

1113 **Lemma 1** (Isometry under exact orthogonality). *Suppose  $k = d$  and  $D^\top D = DD^\top = I_d$ . Then*  
1114  *$D$  is an isometry:  $\|Dv\|_2 = \|v\|_2$  for all  $v \in \mathbb{R}^d$ . Consequently, for any probability measures  $\mu, \nu$*   
1115 *on  $\mathbb{R}^d$  with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

1116 *Proof.* Orthogonality of  $D$  implies preservation of the Euclidean norm, and push-forward commutes  
1117 with the map inside the  $\mathcal{W}_2$  infimum; the integrand is unchanged, so the infimum value is identical.  $\square$   
1118

1119 This case, however, is quite unrealistic as SAE usually rely on the overcompleteness to extract  
1120 meaningful and interpretable concepts. In the overcomplete case,  $D$  is no longer orthonormal, but  
1121 we can still have column-orthonormal dictionary. We will use that to show that we can bound using  
1122 the extremal singular value of  $D^\top D$ .

1123 **Theorem 6** (FID under column orthogonal embeddings). *Let  $D \in \mathbb{R}^{k \times d}$  satisfy  $D^\top D = I_d$  and*  
1124 *denote by  $0 < \sigma_{\min} \leq \sigma_{\max}$  the extreme singular values of  $DD^\top$ . Given two data matrices*  
1125  *$A, A' \in \mathbb{R}^{n \times d}$  (rows are sample vectors), set  $Z = AD^\top \in \mathbb{R}^{n \times k}$  and  $Z' = A'D^\top \in \mathbb{R}^{n \times k}$ . Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

1126 *Proof.* Write  $\mu$  for the empirical measure of  $A$  and  $\nu$  for that of  $A'$ , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

1127 For any coupling  $\pi \in \Pi(\mu, \nu)$  (i.e. a probability measure on  $\mathbb{R}^d \times \mathbb{R}^d$  with marginals  $\mu, \nu$ ) we have,  
1128 by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

1129 Integrating with respect to an arbitrary coupling  $\pi \in \Pi(\mu, \nu)$  yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

1130 The middle integral is exactly the transport cost of the pushed-forward coupling  $(D \times D)_\# \pi$  between  
1131  $\mu_D := D_\# \mu$  and  $\nu_D := D_\# \nu$ . Because the inequalities hold for every  $\pi$ , they hold in particular



1132 for the optimal couplings attaining  $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$  and  $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$ , though these two optima need not  
 1133 coincide. Taking the infimum over  $\boldsymbol{\pi}$  term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1134 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1135 Recognising  $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$  for the Gaussian surrogate and plugging in  $(\mathbf{A}, \mathbf{A}')$  (resp.  $(\mathbf{Z}, \mathbf{Z}')$ )  
 1136 finishes the proof.  $\square$

1137 Essentially, theorem 6 tells us that applying a column-orthogonal overcomplete SAE dictionary  
 1138 cannot distort Fréchet Inception Distance by more than the square of its extremal singular values.  
 1139 When  $D$  is nearly orthogonal – empirically we usually found that  $\sigma_{\min}, \sigma_{\max} \approx 1$  – the result implies  
 1140 that FID measured in the SAE feature space is essentially close to the canonical FID.  $\square$

1141